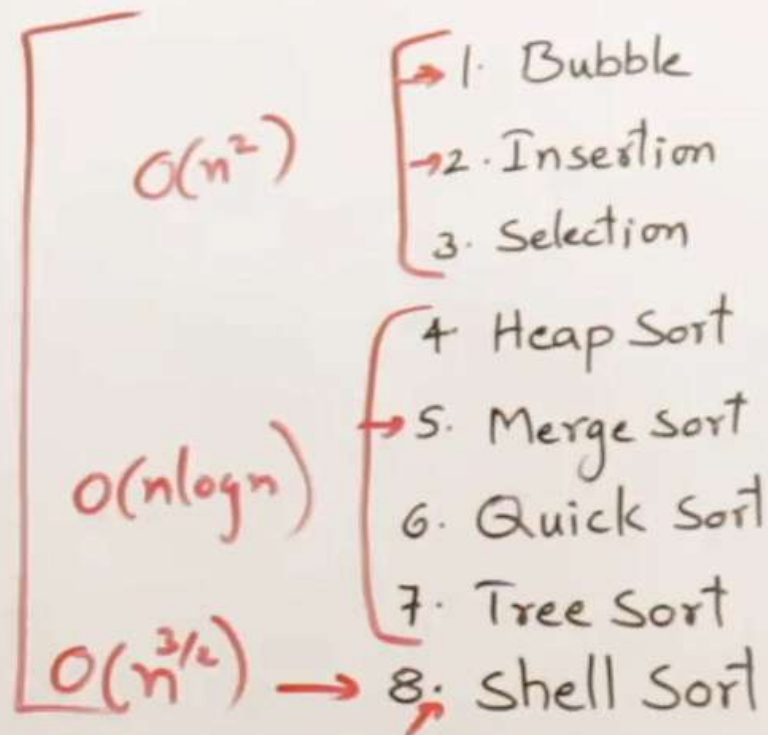
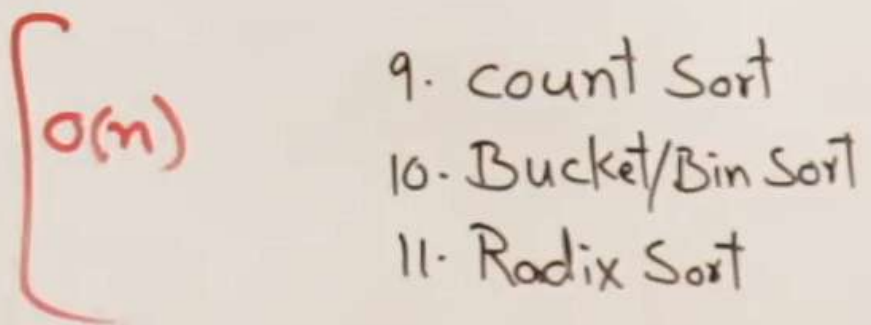


Comparison based Sorts



Index based Sorts



Bubble Sort

$n=5$

A

8	5	7	3	2
0	1	2	3	4

1st Pass

8	5	5	5	5
5	8	7	7	7
7	7	8	3	3
3	3	3	8	2
2	2	2	2	8

4 Comp
4 Swap

2nd Pass

5	5	5	5	
7	7	3	3	
3	3	7	2	
2	2	2	7	8
8	8	8	8	

3 Comp
3 Swap

3rd Pass

5	3	3		
3	5	2		
2	2	5		
7	7	7	8	
8	8	8		

2 Comp
2 Swap

4th Pass

3	2			
2	3			
5	5			
7	7			
8	8			

1 Comp
1 Swap

No. of passes: $4 - (n-1)$ passes

No. of Comp: $1+2+3+4 \rightarrow 1+2+3+\dots+n-1 = \frac{n(n-1)}{2} = O(n^2)$

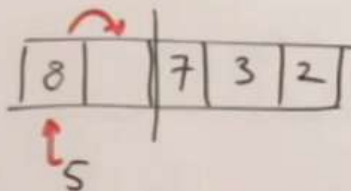
Max No. of Swaps: $1+2+3+4 \rightarrow 1+2+3+\dots+n-1 = \frac{n(n-1)}{2} = O(n^2)$

Insertion Sort

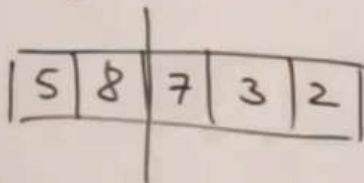
A

8	5	7	3	2
---	---	---	---	---

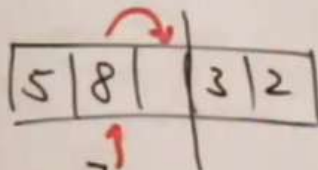
Ist pass Insert 5



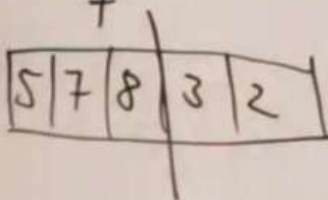
1 comp
1 swap



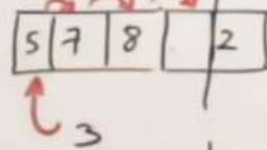
IInd pass Insert 7



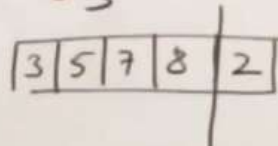
2 comp
2 swap



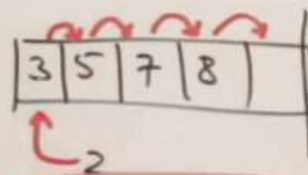
IIIrd pass Insert 3



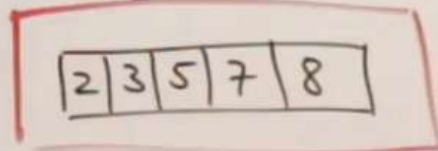
3 comp
3 swap



IVth pass Insert 2



4 comp
4 swap



No. of Passes →

No. of Passes $\rightarrow (n-1)$ passes

No. of Comp $\rightarrow 1+2+3+4+\dots+n-1 = \frac{n(n-1)}{2} = O(n^2)$

No. of swap $\rightarrow 1+2+3+\dots+n-1 = \frac{n(n-1)}{2} = O(n^2)$

```
void insertionSort(int A[], int n)
{
```

```
    for(i=1; i < n; i++)
    {
```

```
        j = i - 1;
```

```
        x = A[i];
```

```
        while(j > -1 && A[j] > x)
        {
```

```
            A[j+1] = A[j];
```

```
            j--;
```

```
        }
```

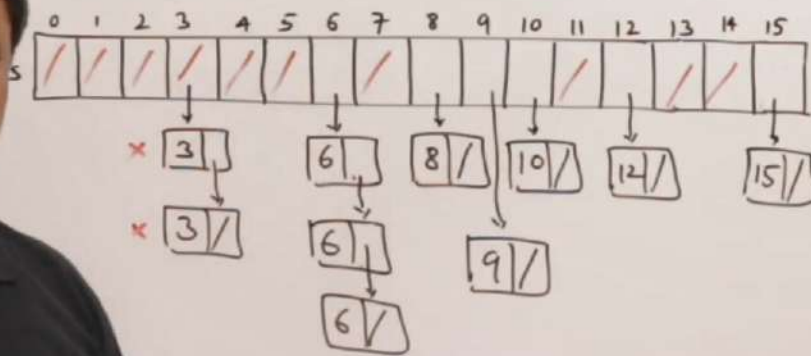
```
        A[j+1] = x;
```

```
    }
}
```

Bucket / Bin Sort

A

3	3	6	6	6	8	9	10	12	15
0	1	2	3	4	5	6	7	8	9



```
void BinSort(int A[], int n)
{
    int max, i, j;
    Node *Bins;
    max = findMax(A, n);
    Bins = new Node[max+1];

    for(i=0; i<max+1; i++)
        Bins[i] = NULL;

    for(i=0; i<n; i++)
    {
        Insert(Bins[A[i]], A[i]);
    }

    i=0; j=0;
    while(i<max+1)
    {
        while(Bins[i] != NULL)
            A[j++] = Delete(Bin(i));
        i++;
    }
}
```

Radix Sort

A	237	146	259	348	152	163	235	48	36	62
	0	1	2	3	4	5	6	7	8	9

$A[i] \% 10$

	0	1	2	3	4	5	6	7	8	9
Bins	/	/			/					
			↓	↓	↓	↓	↓	↓	↓	↓
			152	163	235	146	237	348	259	
			↓			↓		↓		
			62			36		48		

Radix Sort

A	237	146	259	348	152	163	235	48	36	62
	0	1	2	3	4	5	6	7	8	9

152, 62, 163, 235, 146, 36, 237, 348, 48, 259

$$A[i] \neq 10$$
[illegible]

Radix Sort

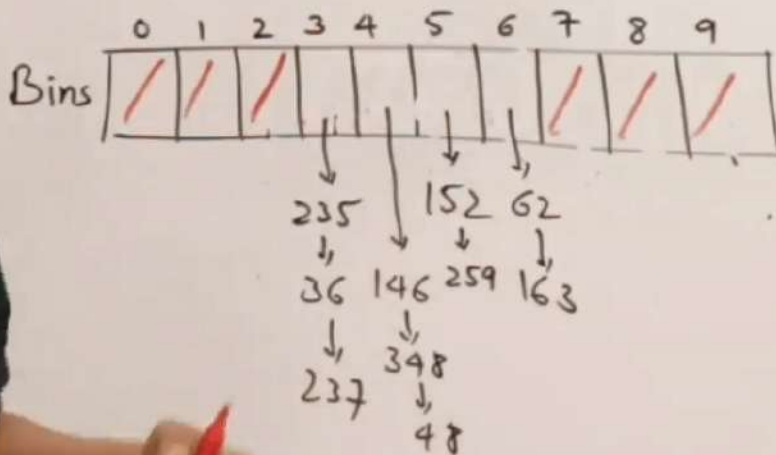
A	237	146	259	348	152	163	235	48	36	62
	0	1	2	3	4	5	6	7	8	9

1st pass

152, 62, 163, 235, 146, 36, 237, 348, 48, 259

$$\textcircled{1} \left\{ (A[i]/1) \% 10 \right.$$

$$\textcircled{2} \left\{ \begin{array}{l} (A[i]/10) \% 10 \\ (152/10) = 15 \% 10 = 5 \end{array} \right.$$



Radix Sort

A	237	146	259	348	152	163	235	48	36	62
	0	1	2	3	4	5	6	7	8	9

1st pass

152, 62, 163, 235, 146, 36, 237, 348, 48, 259

2nd pass

235, 36, 237, 146, 348, 48, 152, 259, 62, 163

	0	1	2	3	4	5	6	7	8	9
Bins	/	/	/	/	/	/	/	/	/	/

$$1. (A[i] / 10) \% 10$$

$$2. (A[i] \% 10)$$

Radix Sort

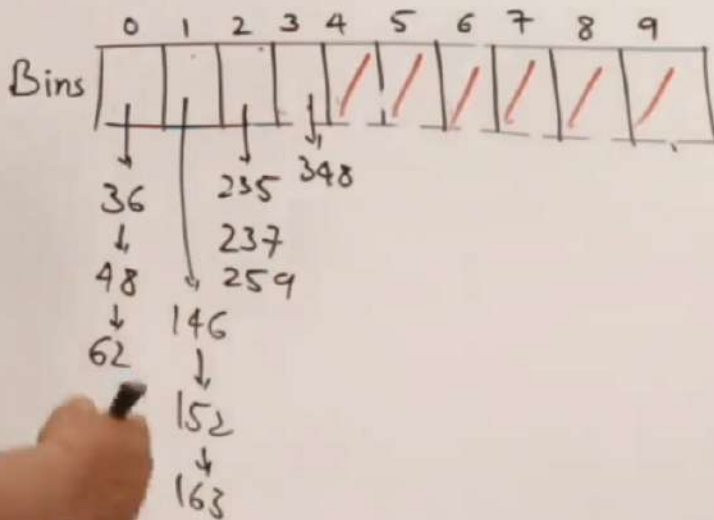
A	237	146	259	348	152	163	235	48	36	62
	0	1	2	3	4	5	6	7	8	9

1st pass

152, 62, 163, 235, 146, 36, 237, 348, 48, 259

2nd pass

235, 036, 237, 146, 348, 048, 152, 259, 062, 163



$$1. (A[i] \% 10) \% 10$$

$$2. (A[i] / 10) \% 10$$

$$3. (A[i] / 100) \% 10$$

$$(235 / 100) \% 10$$

$$2 \% 10 = 2$$

Radix Sort

A	237	146	259	348	152	163	235	48	36	62
	0	1	2	3	4	5	6	7	8	9

ith pass

152, 62, 163, 235, 146, 36, 237, 348, 48, 259

iind pass

235, 36, 237, 146, 348, 48, 152, 259, 62, 163

iiird pass

36, 48, 62, 146, 152, 163, 235, 237, 259, 348

	0	1	2	3	4	5	6	7	8	9
Bins					/	/	/	/	/	/

$$1. (A[i] / 1) \% 10$$

$$2. (A[i] / 10) \% 10$$

$$3. (A[i] / 100) \% 10$$

$$(235 / 100) \% 10$$

$$2 \% 10 = 2$$